

Hypomyelinating Leukodystrophy 23 (HLD23) — Comprehensive Disease Characterization

MONDO: MONDO:0030514 | **OMIM:** 619688 | **Gene:** *RNF220* (HGNC:25552) | **Inheritance:** Autosomal recessive | **Category:** Mendelian

Summary (Answer to the Research Question)

Hypomyelinating Leukodystrophy 23 (HLD23) is an ultra-rare autosomal-recessive **laminopathy** caused by biallelic missense variants in *RNF220*, which encodes a RING-type ubiquitin E3 ligase. It presents as a progressive **hypomyelinating leukodystrophy with cerebellar ataxia and sensorineural deafness**, frequently accompanied by **fibrotic cardiomyopathy and hepatopathy**. Mechanistically, RNF220 dysfunction converges on two axes: (1) reduced binding to **lamin B1**, disrupting nuclear-envelope integrity/morphology (the laminopathy axis), and (2) loss of **K63-linked polyubiquitination/stabilization of Olig1/2** and dysregulated **Shh/Gli, Wnt/ β -catenin** and **IFN-STAT1** signaling, impairing oligodendrocyte development and myelination (the developmental-signaling axis). The disease was first delineated by Sferra et al. (Brain, 2021;

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in seven consanguineous families. There is no disease-specific therapy; management is supportive.

Evidence base: The disease-level knowledge derives almost entirely from one landmark human clinical/molecular study

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plus a substantial body of model-organism and cell-biology work on RNF220. This is aggregated disease-level knowledge (case series + functional studies), not EHR/individual-patient registry data.

1. Disease Information

Overview. HLD23 is a genetically determined disorder of central nervous system white matter (a leukodystrophy) in which myelin is deficient from the outset (hypomyelination) rather than being formed and then lost. The primary lesion is in glia/myelin, with secondary axonal degeneration emerging as the disease progresses. It is distinctive among hypomyelinating leukodystrophies for combining CNS hypomyelination with **cerebellar ataxia, sensorineural deafness, cardiac (fibrotic cardiomyopathy) and hepatic involvement**, and for being mechanistically a **laminopathy**.

"Leukodystrophies are a heterogeneous group of rare inherited disorders that mostly involve the white matter of the CNS. These conditions are characterized by primary glial cell and myelin sheath pathology of variable aetiology, which causes secondary axonal degeneration, generally emerging with disease progression." —

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Key identifiers - MONDO: MONDO:0030514 - **OMIM (phenotype):** #619688 (Leukodystrophy, hypomyelinating, 23, with ataxia, deafness, liver dysfunction, and dilated cardiomyopathy) - **Gene OMIM:** *RNF220* 616136 - **HGNC:** *RNF220* (HGNC:25552); **NCBI Gene:** 55182; **Ensembl:** ENSG00000187147; **UniProt:** Q5T0D9 - **Cytogenetic locus:** 1p34.1 - **Orphanet:** Ultra-rare hypomyelinating leukodystrophy (no dedicated high-prevalence ORPHA entry; subsumed under genetic hypomyelinating leukodystrophies) - **ICD-11:** 8A44.0 (Leukodystrophy) / **ICD-10:** E75.2–G37.x (leukodystrophy, unspecified) - **MeSH:** Hereditary Central Nervous System Demyelinating Diseases / Leukodystrophy

Synonyms / alternative names - HLD23 - Leukodystrophy, hypomyelinating, 23, with ataxia, deafness, liver dysfunction and dilated cardiomyopathy - *RNF220*-related leukodystrophy / *RNF220*-related laminopathy

2. Etiology

Disease causal factors — genetic. HLD23 is monogenic and Mendelian. It is caused by **biallelic (homozygous) missense variants in *RNF220***. Two recurrent variants at adjacent, highly conserved arginine residues were identified (protein-level nomenclature as reported; exact cDNA numbering is transcript-dependent): - **p.(Arg363Gln) [p.R363Q]** - **p.(Arg365Gln) [p.R365Q]**

"We report these two homozygous missense variants (p.R363Q and p.R365Q) in the ubiquitin E3 ligase *RNF220* as the underlying cause of this novel form of leukodystrophy with ataxia and sensorineural deafness that includes fibrotic cardiomyopathy and hepatopathy as associated features in seven consanguineous families." —

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Genetic risk factors. - Causal variants: homozygous *RNF220* p.R363Q / p.R365Q (functional class: hypomorphic/loss-of-function with respect to lamin B1 binding — see §6). - **Susceptibility/modifier loci:** none established. The cofactor **ZC4H2** stabilizes *RNF220* (

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and is a plausible biological modifier, though not demonstrated as a clinical modifier in HLD23.

- **Family history / consanguinity:** parental consanguinity is a major risk factor — all reported families were consanguineous.

Environmental risk factors. None identified; the disorder is fully genetically determined. Sex, toxins, lifestyle and occupational exposures are not implicated.

Protective factors. None described (genetic or environmental). Not applicable to a monogenic recessive disorder beyond the trivial protection of carrying at least one wild-type allele (carriers are unaffected).

Gene–environment interactions. None documented; not applicable.

3. Phenotypes

All frequencies are qualitative given the small published cohort (seven consanguineous families,

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Onset is early (infancy/childhood); course is **progressive**.

Phenotype	Type	HPO term	Onset	Severity	Progression	Frequency
Hypomyelinating leukodystrophy / CNS hypomyelination	Imaging/ clinical sign	HP:0002500 (abnormal cerebral white matter), Leukodystrophy HP:0002415	Infancy/ childhood	Severe	Progressive	Universal (defining)
Cerebellar ataxia	Clinical sign	HP:0001251	Childhood	Moderate–severe	Progressive	Characteristic frequent
Sensorineural hearing loss	Clinical sign / lab (audiometry)	HP:0000407	Childhood	Variable	Progressive	Characteristic frequent
Dilated/fibrotic cardiomyopathy	Physical manifestation	HP:0001644 (dilated CM) / HP:0001638 (CM)	Variable	Potentially severe	Progressive	Associated feature
Hepatopathy / liver dysfunction	Lab abnormality / sign	HP:0001392 (liver abnormality), HP:0001410 (hepatic failure if severe)	Variable	Variable	Progressive	Associated feature
Motor deterioration / spasticity–hypotonia	Clinical sign	HP:0002493 / HP:0001257 / HP:0001252	Childhood	Variable	Progressive	Common (typical of leukodystrophy)
Developmental delay / cognitive impairment	Behavioral/ cognitive	HP:0001263 / HP:0001249	Childhood	Variable	Progressive	Common

Quality-of-life impact. Combined motor (ataxia, spasticity), sensory (deafness), cardiac and hepatic involvement produces high disability burden with progressive loss of ambulation and communication, need for hearing rehabilitation, and cardiac/hepatic morbidity. No disease-

specific QoL instrument (EQ-5D/SF-36/PROMIS) data exist for HLD23; leukodystrophy-general tools (e.g., GMFC-MLD, CFCS) are applicable by analogy (cf.).

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4. Genetic / Molecular Information

- **Causal gene:** *RNF220* (RING finger protein 220), 1p34.1; OMIM 616136; HGNC:25552; NCBI Gene 55182; UniProt Q5T0D9. Encodes an evolutionarily conserved RING-type ubiquitin E3 ligase

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- **Pathogenic variants:** p.(Arg363Gln); p.(Arg365Gln) (cDNA numbering transcript-dependent). **Type:** missense, affecting adjacent conserved arginines. **Zygosity/origin:** homozygous, germline. **ACMG classification:** pathogenic/likely pathogenic (supported by functional co-IP and model-organism data). **Allele frequency:** absent/ultra-rare in gnomAD (consistent with severe recessive disease). **Functional consequence:** partial loss of function — both mutants show **reduced binding to lamin B1**

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the substitutions likely impair substrate/partner interaction rather than abolishing the RING catalytic core.

- **Modifier genes:** none clinically validated. **ZC4H2** (RNF220 stabilizing cofactor;

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and **RLIM**

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are functional partners; ZC4H2 mutations cause a separate human neurodevelopmental disorder.

- **Epigenetic information:** RNF220 acts partly through **EED (PRC2)**, altering histone modification marks at Shh target promoters

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— an epigenetic mechanism at the molecular level, but no disease-specific DNA-methylation signature is reported for HLD23 patients.

- **Chromosomal abnormalities:** none; HLD23 is a single-nucleotide/missense disorder, not a copy-number/structural disorder.

5. Environmental Information

Not applicable. HLD23 is a monogenic autosomal-recessive disorder with **no environmental, lifestyle, toxic, or infectious contributing factors**. The only non-genetic determinant is **consanguinity**, which raises the probability of homozygosity for the recessive allele.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *RNF220* missense mutation (p.R363Q / p.R365Q)** results in a structurally intact but functionally impaired RING E3 ligase.
2. This leads to **reduced binding of RNF220 to lamin B1** (demonstrated by mass-spec + co-IP, [P 33964137](#)).
3. Reduced RNF220–lamin B1 regulation results in **abnormal lamin B1 localization/aggregation and loss of nuclear-envelope/nuclear morphology maintenance** (demonstrated in cells; *Drosophila* lamin Dm0 aggregation with a neurodegenerative phenotype — model-organism evidence, [P 33964137](#)).
→ **Laminopathy branch.**
4. **In parallel (developmental-signaling branch):** impaired RNF220 function reduces **K63-linked polyubiquitination and stabilization of Olig1/2** (mouse, [P 38324685](#)), and dysregulates **Shh/Gli** (via Gli nuclear export and EED/PRC2) and **Wnt/β-catenin** and **IFN-STAT1** signaling

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5. Loss of Olig1/2 stabilization *impedes* **oligodendrocyte progenitor cell (OPC) proliferation, differentiation and (re)myelination** (mouse,

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6. Failed myelination + nuclear-envelope pathology in glia *results in* **CNS hypomyelination (leukodystrophy)** — the defining lesion — with **secondary axonal degeneration** as disease progresses

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7. Dysregulated Shh/Gli and cerebellar development, plus white-matter/cerebellar pathology, *lead to* **cerebellar ataxia** (inferred link, supported by RNF220's demonstrated role in cerebellar development,

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8. Nuclear-envelope/lamin B1 pathology in cochlear, cardiac and hepatic cells *leads to* **sensorineural deafness, fibrotic cardiomyopathy and hepatopathy** (multisystem laminopathy; mechanism inferred from lamin B1's ubiquitous nuclear-lamina role and the observed clinical co-occurrence,

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Detail by category

- **Molecular pathways:** ubiquitin–proteasome/ubiquitin-signaling (K63 chains), Sonic hedgehog (Shh)–Gli, Wnt/β-catenin, IFN–STAT1, PRC2/EED-mediated epigenetic regulation.
- **Cellular processes:** oligodendrocyte differentiation and myelination; neural stem/progenitor proliferation vs differentiation balance (loss of RNF220 promotes premature neuronal differentiation,

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nuclear-envelope maintenance; secondary axonal degeneration.

- **Protein dysfunction:** partial loss-of-function of the E3 ligase; **loss of a protein–protein interaction (RNF220–lamin B1)** and **failure to stabilize substrates (Olig1/2, Gli, β -catenin, STAT1)**; downstream **lamin B1 mislocalization/aggregation**.

- **Metabolic/biochemical:** the core defect is enzymatic (ubiquitin-ligase) rather than metabolic; no primary metabolite abnormality. Myelin lipid deficiency is secondary to hypomyelination.

- **Immune involvement:** RNF220 modulates IFN–STAT1 signaling

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a neuroinflammatory component is plausible (as in other leukodystrophies) but not specifically characterized in HLD23.

- **Tissue damage mechanisms:** nuclear-envelope destabilization → cellular dysfunction/degeneration; myelin deficiency → axonal vulnerability; cardiac/hepatic **fibrosis**.

- **Suggested ontology terms:** GO:0004842 ubiquitin-protein transferase activity; GO:0070534 protein K63-linked ubiquitination; GO:0042552 myelination; GO:0048709 oligodendrocyte differentiation; GO:0007224 smoothened/Shh signaling; GO:0016055 Wnt signaling; GO:0060333 IFN- γ -mediated signaling. **Cellular components:** GO:0005635 nuclear envelope; GO:0005638 nuclear lamina. **Cell types (CL):** CL:0000128 oligodendrocyte; CL:0002453 oligodendrocyte precursor cell; CL:0000127 astrocyte; CL:0000540 neuron.

7. Anatomical Structures Affected

- **Organ level (primary):** brain — cerebral and cerebellar white matter (CNS). **Body system:** nervous system (central).

- **Secondary organ involvement:** inner ear/cochlea (sensorineural deafness); heart (fibrotic/dilated cardiomyopathy — cardiovascular system); liver (hepatopathy — digestive/hepatobiliary system).

- **Tissue/cell level:** nervous tissue — **oligodendrocytes (CL:0000128) and OPCs (CL:0002453)** are the primary affected cells; secondary axonal (neuronal) degeneration; cardiac myocytes and hepatocytes affected in a fibrotic pattern.

- **Subcellular level:** **nuclear envelope (GO:0005635)** and **nuclear lamina (GO:0005638)** — the defining subcellular compartment in this laminopathy; the nucleus broadly.

- **Localization (UBERON):** cerebral white matter UBERON:0002316; cerebellum UBERON:0002037; corpus callosum UBERON:0002336; cochlea/inner ear UBERON:0001844; heart UBERON:0000948; liver UBERON:0002107. **Lateralization:** bilateral/symmetric (typical of hypomyelinating leukodystrophy).
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8. Temporal Development

- **Onset:** early — infancy to childhood; insidious/chronic onset (hypomyelination is present from early development).
 - **Progression:** chronic and **progressive**, with motor deterioration and secondary axonal degeneration accruing over time; multisystem features (cardiac, hepatic) may evolve.
 - **Course pattern:** progressive/neurodegenerative rather than relapsing-remitting or episodic; lifelong.
 - **Critical periods:** early myelination window (infancy/early childhood) is the period of greatest vulnerability and the theoretical window for any future myelin-directed intervention.
 - **Remission:** none; no spontaneous remission described.
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9. Inheritance and Population

- **Inheritance: Autosomal recessive** (homozygous variants in all reported families).
- **Penetrance:** presumed complete in biallelic homozygotes (all reported homozygotes affected).
- **Expressivity:** variable across the multisystem features (degree of cardiac/hepatic involvement varies).
- **Genetic anticipation / germline mosaicism:** not applicable / not reported.
- **Founder effects & consanguinity:** the two recurrent variants in **consanguineous families** indicate consanguinity-driven homozygosity and possible local founder alleles.
- **Carrier frequency:** not established; variants are ultra-rare/absent in gnomAD.
- **Epidemiology: ultra-rare** — only a small number of families reported worldwide; precise prevalence/incidence unknown (not quantified in Orphanet/GBD).

- **Population demographics:** reported in consanguineous pedigrees; no established sex bias (autosomal); pediatric onset. Geographic distribution follows populations with higher consanguinity rates.

10. Diagnostics

- **Imaging (key):** brain **MRI** shows a **hypomyelination pattern** — mild diffuse T2 hyperintensity with near-normal/mildly reduced T1 signal of white matter, best assessed after ~1–2 years of age. MRI pattern recognition is central to leukodystrophy diagnosis

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MR spectroscopy may be adjunctive

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- **Genetic testing (confirmatory):** trio **whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)**, or a **hypomyelinating-leukodystrophy NGS gene panel** including *RNF220*; single-gene testing if the recurrent variant is suspected in a consanguineous family. WES diagnostic yield in leukodystrophy cohorts ~60%

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Variant interpretation per ACMG/AMP.

"The total diagnostic rate of WES was 60.7%." —

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- **Supporting/organ workup:** audiometry/BAER (sensorineural deafness), echocardiography/ECG and cardiac MRI (cardiomyopathy), liver function tests and hepatic imaging (hepatopathy).
- **Clinical criteria / differential diagnosis:** no formal consensus criteria; diagnosis is gene-based. Differentials include other hypomyelinating leukodystrophies — PLP1 (PMD), GJC2 (HLD2), *TUBB4A* (H-ABC/HLD6), POLR3-related (4H) leukodystrophy, *FAM126A*/HYCC1 (HLD5), and other laminopathies; distinguishing features here are the **combination of hypomyelination + ataxia + sensorineural deafness + cardiomyopathy + hepatopathy** and the *RNF220* genotype.

- **Screening:** no newborn screening exists (no biochemical marker); **cascade/carrier testing** in affected consanguineous families is appropriate.

11. Outcome / Prognosis

- **Survival/mortality:** disease-specific survival data are not established given the few reported patients. Prognosis is guarded due to progressive neurodegeneration plus potentially life-limiting **cardiomyopathy** and **hepatopathy**.
- **Morbidity/disability:** high — progressive motor disability (ataxia, spasticity), sensory loss (deafness), and cardiac/hepatic morbidity; likely loss of independent ambulation and communication over time (by analogy to other progressive leukodystrophies,

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- **Recovery potential:** none without disease-modifying therapy; hypomyelination is not spontaneously reversible.
- **Prognostic factors:** extent of cardiac/hepatic involvement and rate of neurological progression are the principal determinants; specific molecular prognostic biomarkers are not defined.

12. Treatment

No disease-specific or curative therapy exists for RNF220-related HLD23. Unlike enzyme-deficiency leukodystrophies with approved disease-modifying options (e.g., **HSCT** for Krabbe disease,

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41604001; **arsa-cel gene therapy** for metachromatic leukodystrophy,

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RNF220-HLD23 has no such intervention. Care is **multidisciplinary and supportive/symptomatic:**

- **Supportive care** (NCIT:C15277): symptom management, nutrition, spasticity/ataxia management.

- **Rehabilitation:** physical therapy (NCIT:C15535), occupational therapy, speech/communication therapy.
 - **Sensory:** hearing aids / **cochlear implantation** for sensorineural deafness.
 - **Cardiac:** surveillance and standard heart-failure/cardiomyopathy management; consider device/transplant evaluation per cardiology.
 - **Hepatic:** monitoring and supportive hepatic care.
 - **Pharmacotherapy:** only symptomatic (e.g., anti-spasticity agents, anticonvulsants if seizures); **no pharmacogenomic** guidance specific to HLD23.
 - **Advanced/experimental therapeutics:** none in clinical trials for HLD23 specifically; gene-replacement/genome-editing and small-molecule myelin-repair strategies are conceptual future directions. No NCT identifiers for RNF220-HLD23.
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13. Prevention

- **Primary prevention:** not possible for disease occurrence beyond reproductive genetic measures; no vaccine/lifestyle modification (non-environmental disease).
 - **Genetic prevention/counseling** (principal lever): **genetic counseling** for autosomal-recessive 25% recurrence risk; **carrier screening** in consanguineous families/populations; **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** for known familial variants; cascade testing of relatives.
 - **Secondary prevention:** early identification of at-risk siblings via cascade testing; early audiologic, cardiac and hepatic surveillance to manage complications.
 - **Tertiary prevention:** proactive cardiac and hepatic monitoring, rehabilitation, and management of contractures/scoliosis to limit complications.
 - **Public-health note:** in high-consanguinity populations, community genetic counseling reduces incidence of recessive disorders generally.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *RNF220* is evolutionarily conserved

([P 34716995](#)).

Orthologs: **mouse** *Rnf220* (NCBI Gene 66743; NCBITaxon:10090), **zebrafish** *rnf220a/rnf220b* (NCBITaxon:7955), **Drosophila** functional counterpart acting on **lamin Dm0** (NCBITaxon:7227).

- **Natural disease in other species:** no naturally occurring *RNF220*-associated disease is documented in companion animals or wildlife (no OMIA entry known); relevance is experimental, not veterinary.

- **Comparative biology:** the **RNF220–lamin B1** interaction is conserved to *Drosophila* (*RNF220* knockdown → lamin Dm0 aggregation and neurodegeneration,

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demonstrating deep evolutionary conservation of the disease mechanism.

- **Transmission:** not applicable (non-infectious, non-zoonotic).

15. Model Organisms

- **Mouse (*Mus musculus*):**

- **Rnf220 knock-in model of the leukodystrophy-related variant and conditional/oligodendrocyte-lineage depletion models:** recapitulate impaired OPC proliferation/differentiation and (re)myelination with learning/memory deficits — good recapitulation of the **hypomyelination** component

([P 38324685](#)).

- **Cerebellar development / Shh-medulloblastoma models**

([P 32376680](#))

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— relevant to the **ataxia**/cerebellar axis.

- Noradrenergic (locus coeruleus, Phox2 monoubiquitylation) and motor-neuron development models
([P 32094113](#)).
- **Zebrafish (*Danio rerio*):** neural-patterning models of RNF220/Shh (reviewed
([P 34716995](#)).
- **Drosophila melanogaster:** RNF220 silencing → lamin Dm0 mislocalization/aggregation and neurodegeneration — models the **laminopathy** axis
([P 33964137](#)).
- **In vitro / cellular:** patient-mutant co-IP (reduced lamin B1 binding); neural stem cell (NSC) studies showing loss of RNF220/ZC4H2 inhibits proliferation and promotes differentiation
([P 32630355](#));
myoblast differentiation studies of ΔN-RNF220 isoform
([P 40609864](#)).
- **Genetic model types available:** knock-in, conditional/lineage-specific knockdown/knockout (mouse); morphant/mutant (zebrafish); RNAi knockdown (Drosophila).
- **Phenotype recapitulation & limitations:** mouse models reproduce myelination defects and cerebellar/Shh phenotypes; Drosophila reproduces the lamin B1 axis. **Limitation:** no single published model simultaneously recapitulates the **full human multisystem phenotype** (leukodystrophy + deafness + cardiomyopathy + hepatopathy).
- **Resources:** MGI (*Rnf220*), ZFIN (*rnf220a/b*), FlyBase, Alliance of Genome Resources.

Supported vs. Refuted Hypotheses

Supported 1. HLD23 is caused by biallelic *RNF220* missense variants (p.R363Q/p.R365Q), AR inheritance

([P 33964137](#)).

2. HLD23 is a **laminopathy**: mutant RNF220 shows reduced lamin B1 binding; RNF220 maintains nuclear morphology

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3. The hypomyelination arises from failed RNF220-mediated K63-ubiquitination/stabilization of Olig1/2 and disrupted oligodendrocyte development

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4. RNF220 acts pleiotropically via Shh/Gli, Wnt, IFN-STAT1, explaining multisystem/cerebellar features

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Refuted / excluded - Environmental, infectious, metabolic-primary, or chromosomal/structural causation — excluded; the disorder is a single-gene missense laminopathy. - Existence of an approved disease-modifying therapy — none exists (management is supportive).

Limitations and Future Directions

- **Small evidence base:** core clinical knowledge rests on one case series (seven consanguineous families,

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prevalence, natural history, penetrance quantification, sex/age distributions, and QoL metrics are not established.

- **Mechanistic gaps:** the causal steps linking lamin B1 dysfunction to deafness, cardiomyopathy and hepatopathy are **inferred**, not directly demonstrated; the relative contribution of the lamin-B1 axis vs. the Olig1/2/Shh axis to human hypomyelination is unquantified.
- **Therapeutic gap:** no targeted therapy or clinical trial; future work could explore genotype-specific interventions, myelin-repair agents, and a model that captures the full multisystem phenotype.
- **Ontology mapping:** MONDO:0030514 / OMIM 619688 / *RNF220* (HGNC:25552); HPO, GO, CL, UBERON, NCIT terms suggested throughout.

Key References (PMID)

- **33964137** — Sferra et al., *Brain* 2021. Landmark: biallelic *RNF220* mutations cause laminopathy with leukodystrophy, ataxia, deafness (+ cardiomyopathy, hepatopathy).
- **38324685** — Li et al., 2024. RNF220 K63-ubiquitinates/stabilizes Olig1/2 in oligodendroglial development and myelination; knock-in leukodystrophy mouse.
- **34716995** — Ma & Mao, 2022. Review: RNF220 in neural development (Shh/Gli, Wnt, IFN-STAT1; ZC4H2).
- **32376680 / 35040952 / 32094113 / 32630355 / 39526890 / 40609864** — RNF220 developmental biology and models.
- **28638987** — van der Knaap & Bugiani, 2017. Leukodystrophy classification; MRI+NGS diagnostic paradigm.
- **37597066** — Iranian Leukodystrophy Registry; WES diagnostic yield 60.7%.
- **40267426 / 42040243 / 41604001** — Contrast: disease-modifying therapies (gene therapy/H SCT) in other leukodystrophies (MLD, Krabbe).

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